Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

ECH617S – MEDICINAL CHEMISTRY

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. The study of the absorption, distribution and metabolism of drugs _____
   a) Pharmacophore
   b) Medicinal chemistry
   c) Pharmacokinetics
   d) Molecular pharmacology

2. SAR stands for
   a) Structure activity relationship
   b) Structure and reactivity
   c) Stereochemistry and reactivity
   d) Stereospecific and reactivity

3. Antibiotics effective against both gram-positive and gram-negative bacteria.
   a) Ampicillin b) Capreomycin
   c) Paromomycin d) All the above

4. Which of the following is referred to as queen of drugs?
   a) Streptomycin b) Sulphadiazene
   c) Pencillin d) Choromycetin

5. Which of the following is a Cardiovascular drug?
   a) Tetramycin b) Streptomycin c) Digitoxin d) Methadone

6. Which is an antineoplastic agent?
   a) Chlorambucil b) Vitamin A
   c) Ripamycin d) None of these

7. Which is not true of sulpha drugs?
   a) Sulphanilamides
   b) Effective chemotherapeutic agent
   c) Antibiotics
   d) Cure of bacterial infections

8. Which one of the following has an azo group?
   a) Sulphadiazine b) Prontosil
   c) Streptomyin d) All the above

9. Barbiturates are derived from
   a) Barbituric acid b) Alcohol c) Aldehyde d) Ketone

10. Buspirone is useful in the treatment of
    a) Fever b) Anxiety c) Psychotic diseases d) None

II. Fill up the blanks

11. _____ are actually macromolecules present in tissues that combine chemically with the drug.

12. _____ was the first effective broad spectrum antibiotic.

13. _____ are used in the treatment of diseases of heart and blood vessels.

14. _____ shown good activity against urinary tract infections.

15. The major use of antipsychotics is to treat _____.

III. Match the following

16. Hansch analysis	- Malignant
17. β-lactum	- Antimicrobial
18. Cancer	- Psychotic depression
19. Sulphonamide	- Penicillin
20. Oxazapam	- Partition coefficient

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. What are drugs? Give an examples.
22. What are drug receptors?
23. What are antibiotics? How are antibiotics classified?
24. Give the therapeutic uses of tetracycline.
25. List out the methods generally used for the treatment of cancer.
26. What are hormones? Give an example.
27. Distinguish between antiseptics and anteinfective drugs.
28. Give the therapeutic uses of chloroquine.
29. Name any two anti-anxiety drugs.
30. Write a note on antidepressants.
31. What are alkylating agents? Give an example.
32. Draw the structure of glutethimide.

SECTION – C (5X7=35)

Answer any **TEN** Questions

33. Explain the rate and induced fit theories of drug activity. **(4+3)**

34. i) Describe the biosynthesis of antibiotics. **(3)**
    ii) Outline the synthesis of chloramphenicol. **(4)**

35. What are antineoplastic drugs? How are they classified? Give an example for each. **(3+4)**

36. i) Explain the mode of action of anti infective drugs. **(4)**
    ii) Give the therapeutic uses of nalidixic acid. **(3)**

37. What are hypnotics and sedatives? Give one example for each. **(3+3+1)**

38. Discuss with examples any two kinds of cardiovascular drugs. **(4+3)**

39. i) Draw the structures of the following compounds. **(2+2+2)**
    i) Amino salicylic acid ii) Diazepam iii) Barbiturates
    ii) What is ampicillin? **(1)**

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